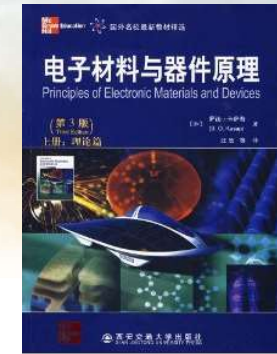
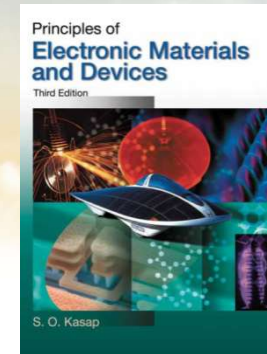




南方科技大学
SOUTHERN UNIVERSITY OF SCIENCE AND TECHNOLOGY

《固态电子学》 Solid-State Electronics



Ch3-4 Elementary Quantum Physics

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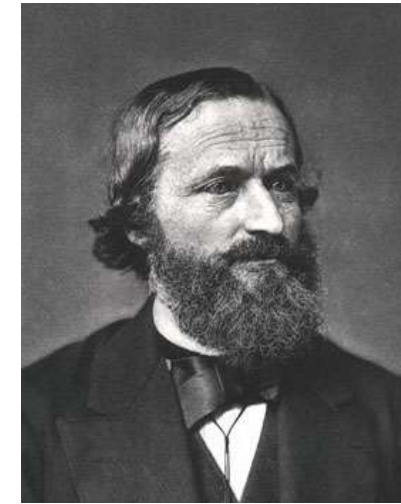


Some slides are modified from other Profs' PPTs. Their contributions are greatly acknowledged.

Dark & White Lines in Spectrum



In 1814, with the help of spectrometers, Fraunhofer observed hundreds of dark lines in sun radiation. He carefully numbered them. In order to commemorate the contribution, people call them the Fraunhofer lines.



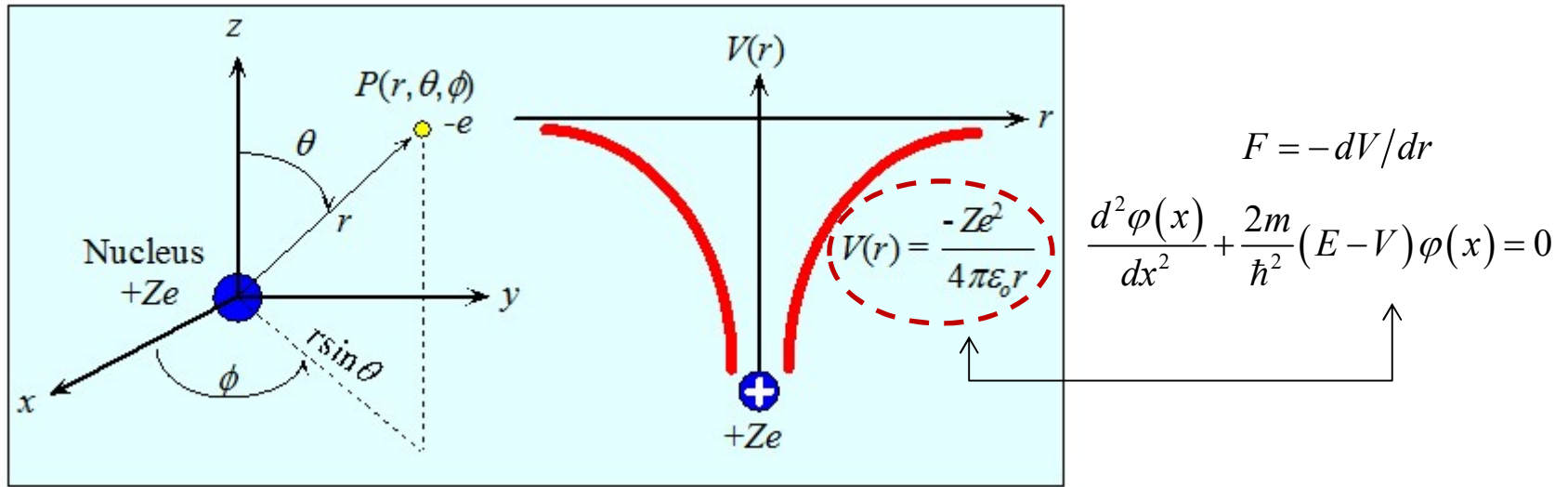
In 1859, Kirchhoff noted that some elements emit bright lines in the spectrum under heating. When continuous light travel through the steam made of this element, a dark line appeared in the same position the same as the bright line.

- The bright line is derived from the emission (emission spectrum)
- The dark line is derived from the absorption (absorption spectrum)

The elements can emit and absorb light, and the two correspond to each other



Hydrogenic Atom: e Wavefunction



- Wave function in spherical coordinates: $\varphi(r, \theta, \phi) = R(r)Y(\theta, \phi)$

$$\varphi(r, \theta, \phi) = \varphi_{n,l,m_l}(r, \theta, \phi) = R_{n,l}(r)Y_{l,m_l}(\theta, \phi) \quad \begin{cases} R_{n,l}(r) & \text{Radial function (径向函数)} \\ Y_{l,m_l}(\theta, \phi) & \text{Spherical harmonic function (球谐函数)} \end{cases}$$

- In an analogy with the 3D potential well, there should be 3 quantum numbers, namely n, l, m_l to characterize the φ, E and p of the electron in this system.
 - 1) Principal quantum number $n = 1, 2, 3, \dots$
 - 2) Orbital angular momentum quantum number $l = 0, 1, 2, \dots (n-1)$
 - 3) Magnetic quantum number $m_l = -l, -(l-1), \dots, 0, \dots, (l-1), l$



Quantum Number

We need to define three quantum numbers to know wavefunction, energy and momentum of the 'e' in a H-atom.

Quantum Number	Symbol	Description	Value
Principal quantum number	n	The number of the shell. Related to size and energy of the orbital (electron)	1, 2, 3 etc. (i.e. any positive integer)
Orbital angular momentum quantum number	l	Related to the shape of the orbit and angular momentum of the electron about the nucleus. It has a small effect on the energy of the electron.	$0, 1, \dots, (n-1)$ $l = 0, 1, 2, 3, \dots$ $s, p, d, f, \dots$
Magnetic quantum number	m_l	Related to orientation of the orbital in space relative to the other orbitals. No effect on the energy of the electron, unless the atom is placed in a magnetic field.	$-l$ to l (including 0)
Spin quantum number	s	Related to the spin states of the electron. It has a very small effect on the electron's energy.	$-1/2$ or $+1/2$



Quantum Number

Table 3.1 Labeling of various $n\ell$ possibilities

		ℓ				
n		0	1	2	3	4
1	K	1s				
2	L	2s	2p			
3	M	3s	3p	3d		
4	N	4s	4p	4d	4f	
5	O	5s	5p	5d	5f	5g

- 1) Principal quantum number $n = 1, 2, 3, \dots$
- 2) Orbital angular momentum quantum number $l = 0, 1, 2, \dots (n-1)$
- 3) Magnetic quantum number $m_l = -l, -(l-1), \dots, 0, \dots, (l-1), l$

s : sharp, p : principal, d : diffuse, f : fundamental, $g, h, i, k, \dots$

它的名称源于对其原子光谱特征谱线外观的描述，分为锐系光谱（**s**harp）、主系光谱（**p**rincipal）、漫系光谱（**d**iffuse）、基系光谱（**f**undamental），其余则依字母序命名（跳过 j ）

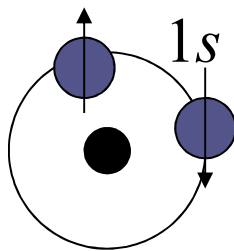
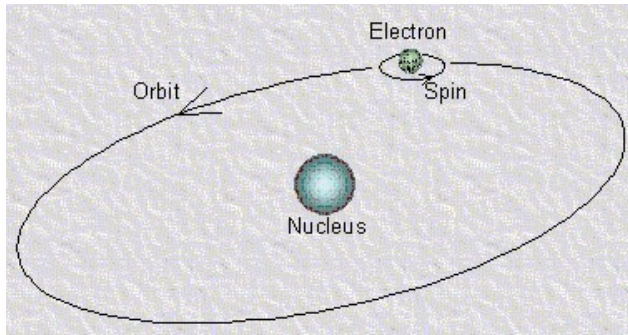


Quantum Number

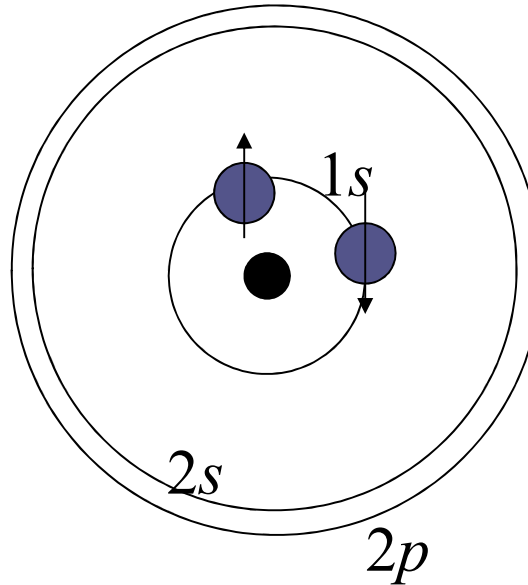
Atomic Orbitals with n up to 4			
n	l	Name	Number of different m_l orbitals
1	0	$1s$	1 ($m_l = 0$)
2	0	$2s$	1 ($m_l = 0$)
	1	$2p$	3 ($m_l = -1, 0, 1$)
3	0	$3s$	1 ($m_l = 0$)
	1	$3p$	3 ($m_l = -1, 0, 1$)
	2	$3d$	5 ($m_l = -2, -1, 0, 1, 2$)
4	0	$4s$	1 ($m_l = 0$)
	1	$4p$	3 ($m_l = -1, 0, 1$)
	2	$4d$	5 ($m_l = -2, -1, 0, 1, 2$)
	3	$4f$	7 ($m_l = -3, -2, -1, 0, 1, 2, 3$)



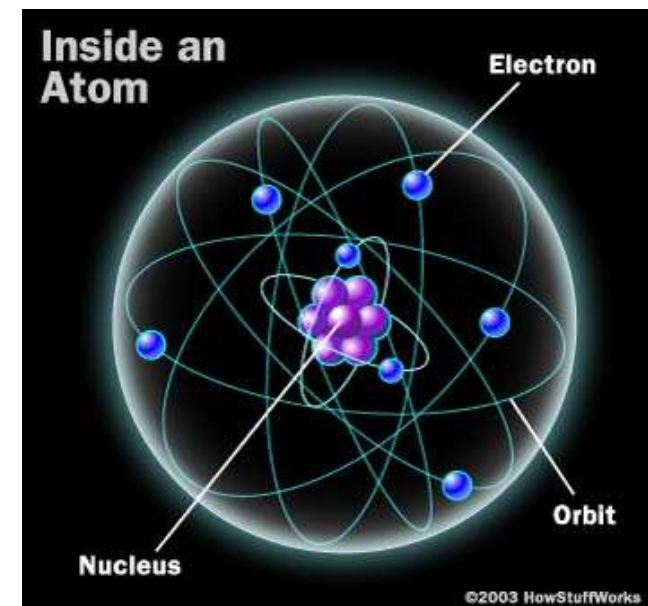
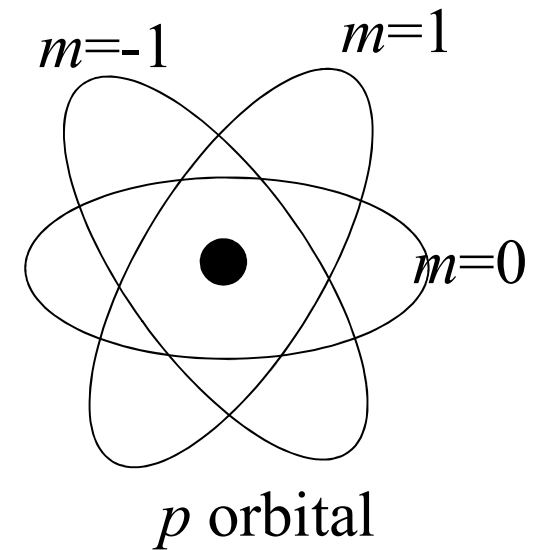
Quantum Number and Orbital



$n=1$
No. of e=2



$n=2$
No. of e=2+6=8



Questions: for a given l sub-orbital, how many electrons can it have?

Answer: $2(2l+1)$



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Hydrogenic Atom: e Wavefunction

Table 3.2 The radial and spherical harmonic parts of the wavefunction in the hydrogen atom ($a_o = 0.0529$ nm)

n	ℓ	$R(r)$	m_ℓ	$Y(\theta, \phi)$
1	0	$\left(\frac{1}{a_o}\right)^{3/2} 2 \exp\left(-\frac{r}{a_o}\right)$	0	$\frac{1}{2\sqrt{\pi}}$
2	0	$\left(\frac{1}{2a_o}\right)^{3/2} \left(2 - \frac{r}{a_o}\right) \exp\left(-\frac{r}{2a_o}\right)$	0	$\frac{1}{2\sqrt{\pi}}$
2	1	$\left(\frac{1}{2a_o}\right)^{3/2} \left(\frac{r}{\sqrt{3}a_o}\right) \exp\left(-\frac{r}{2a_o}\right)$	0	$\frac{1}{2}\sqrt{\frac{3}{\pi}} \cos \theta$
			1	$\frac{1}{2}\sqrt{\frac{3}{2\pi}} \sin \theta e^{j\phi}$
			-1	$\frac{1}{2}\sqrt{\frac{3}{2\pi}} \sin \theta e^{-j\phi}$
				$\left\{ \begin{array}{l} \propto \sin \theta \cos \phi \\ \propto \sin \theta \sin \phi \end{array} \right.$
				Correspond to $m_\ell = -1$ and $+1$.

$$\varphi(r, \theta, \phi) = \varphi_{n,l,m_l}(r, \theta, \phi) = R_{n,l}(r) Y_{l,m_l}(\theta, \phi)$$

The probability of finding the electron at a distance r within a thin spherical shell of radius r and thickness δr

$$\left| \overline{Y_{l,m_l}(\theta, \phi)} R_{n,l}(r) \right|^2 \times (4\pi r^2 \delta r)$$

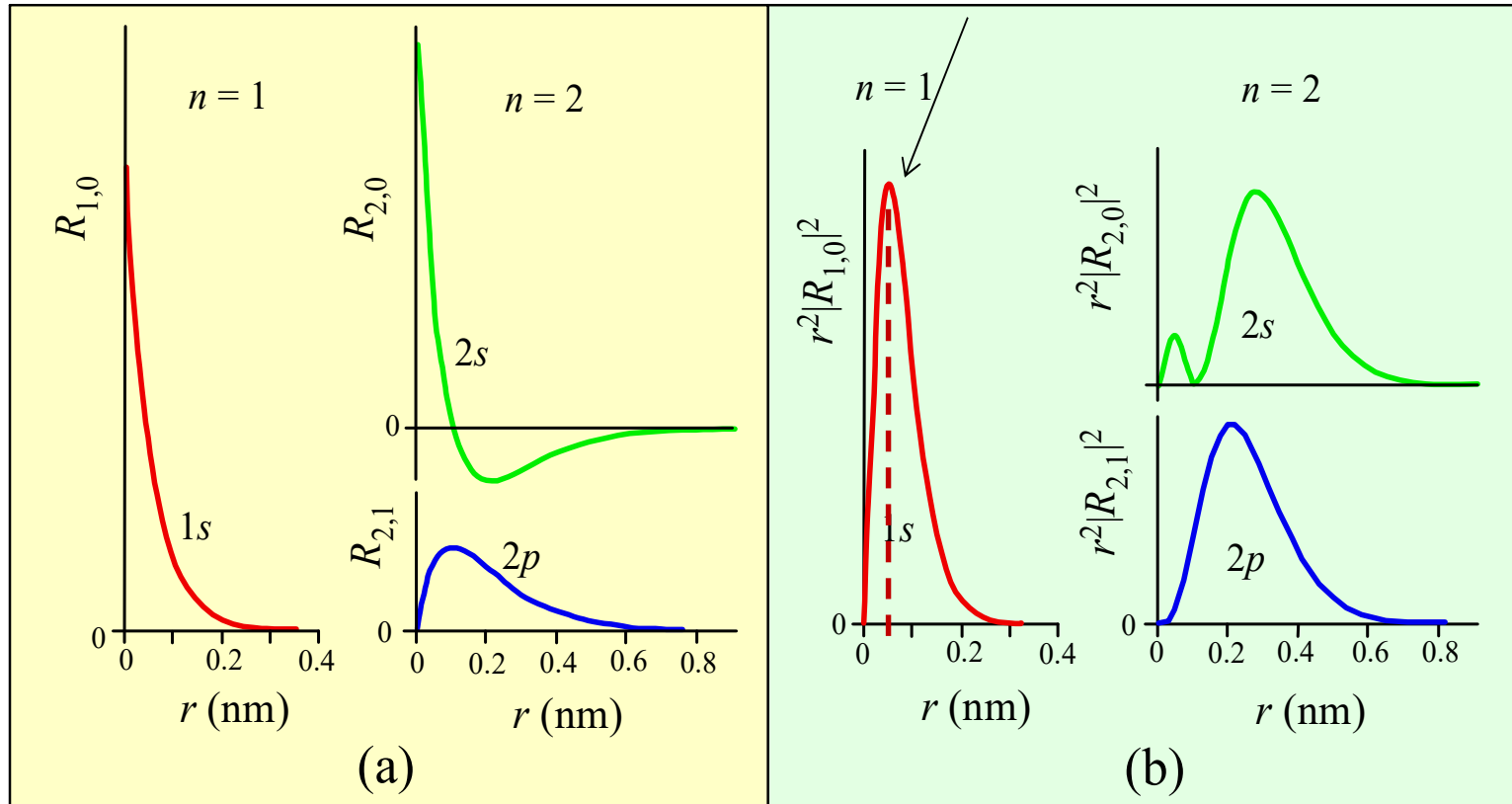
$$\delta P(r) = |R_{n,l}(r)|^2 r^2 \delta r$$

Radial probability density (径向概率密度函数)



e Wavefunction: Radial (R)

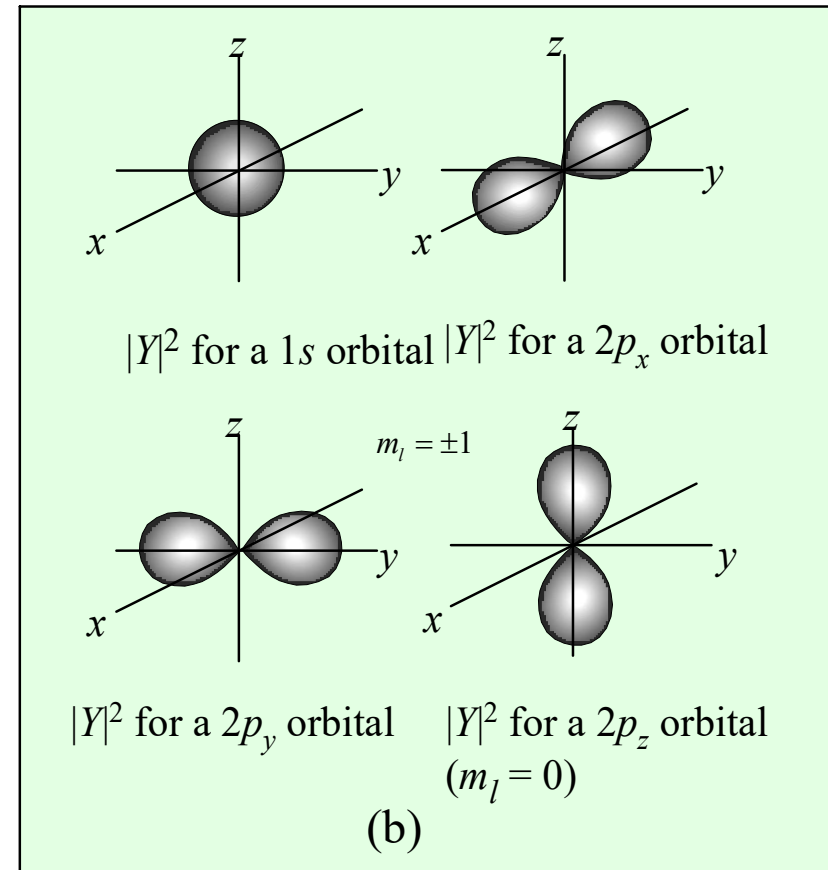
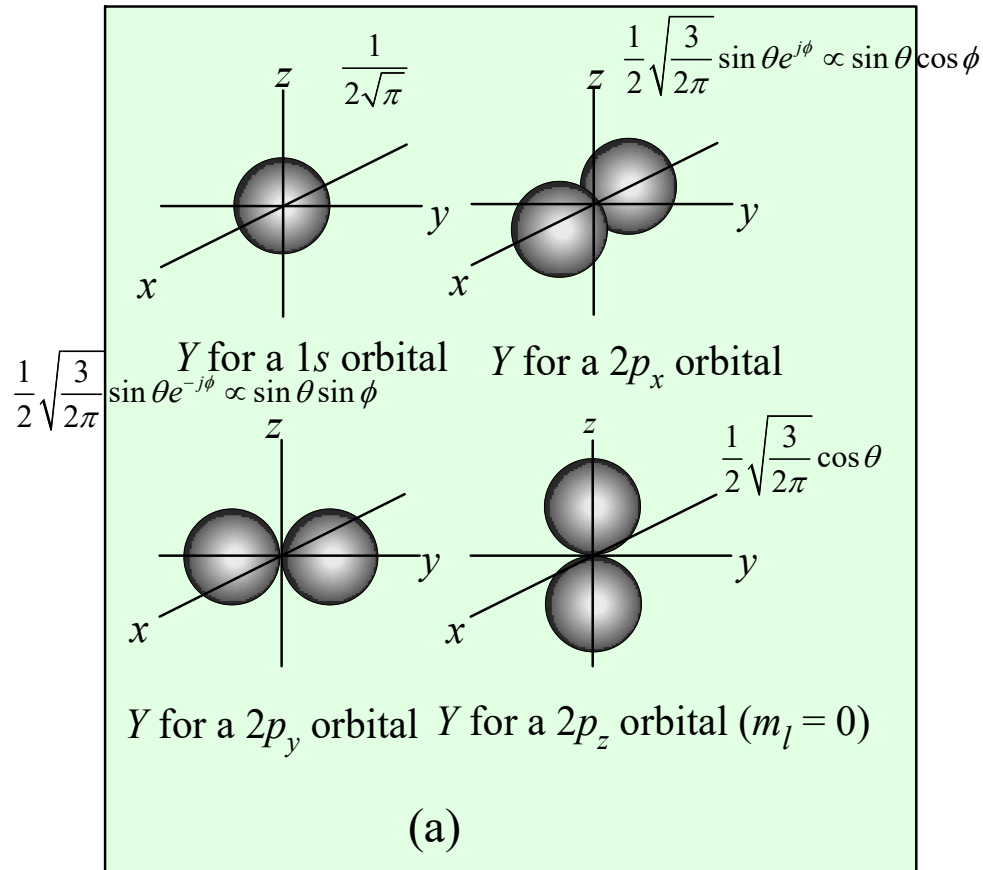
Bohr Radius: $r = a_0 = 0.0529 \text{ nm}$



(a) Radial wavefunctions of the electron in a Hydrogenic atom for various n and l values. (b) the radial probability density. Vertical axis scales are linear in arbitrary units.



e Wavefunction: Spherical Harmonic (Y)

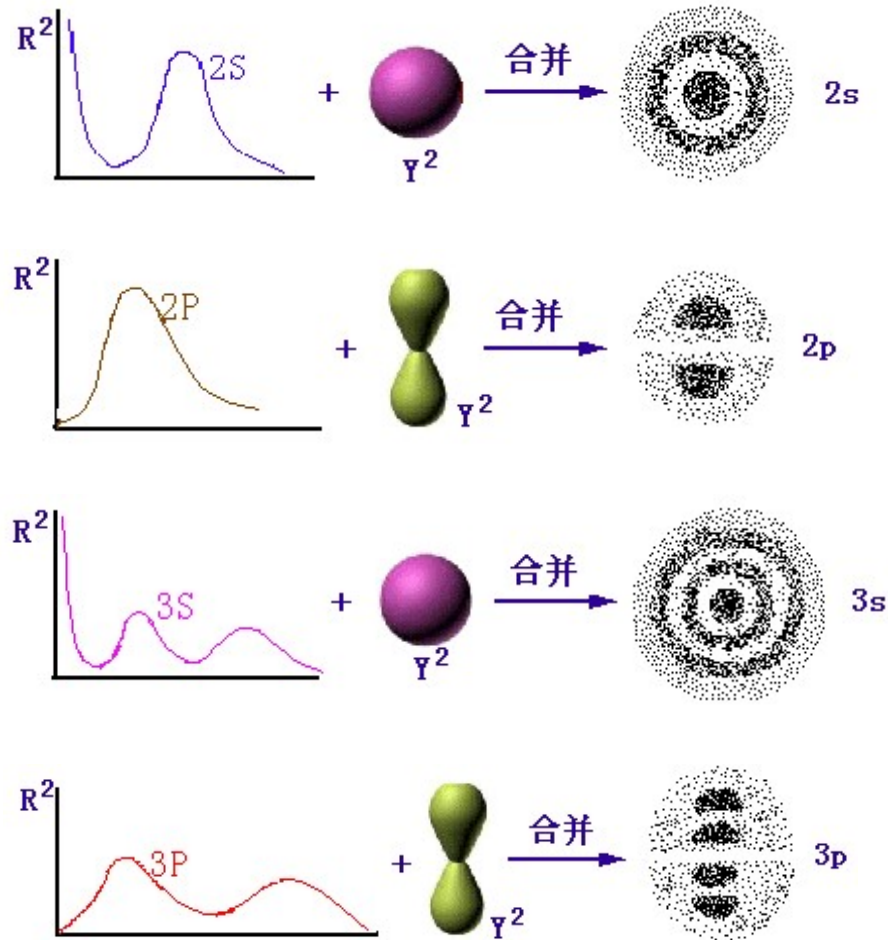


(a) The polar plots of $Y_{n,l}(\theta, \phi)$ for 1s and 2p states. (b) The angular dependence of the probability distribution which is proportional to $|Y_{n,l}(\theta, \phi)|^2$.

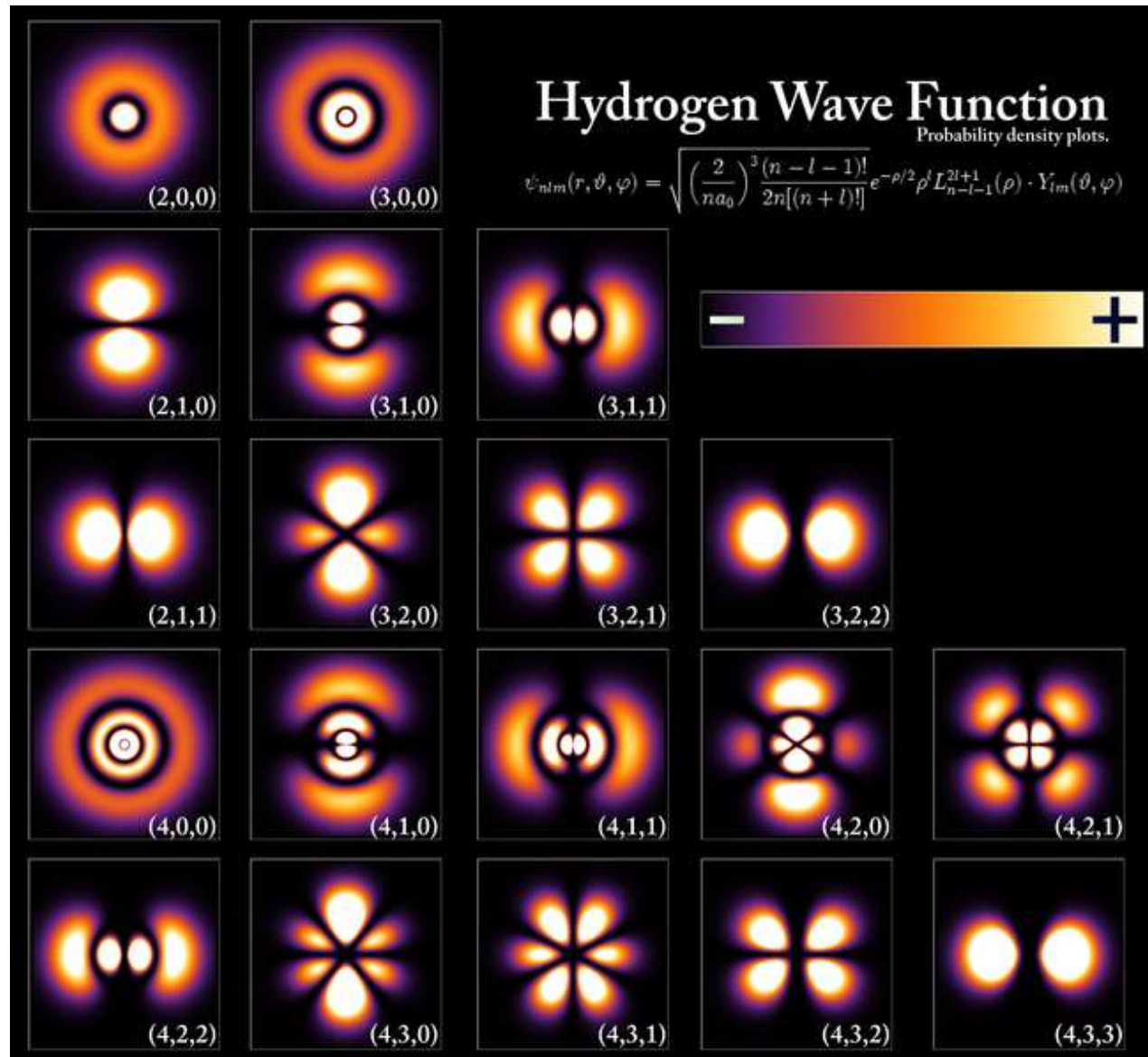


e Orbital: e Cloud

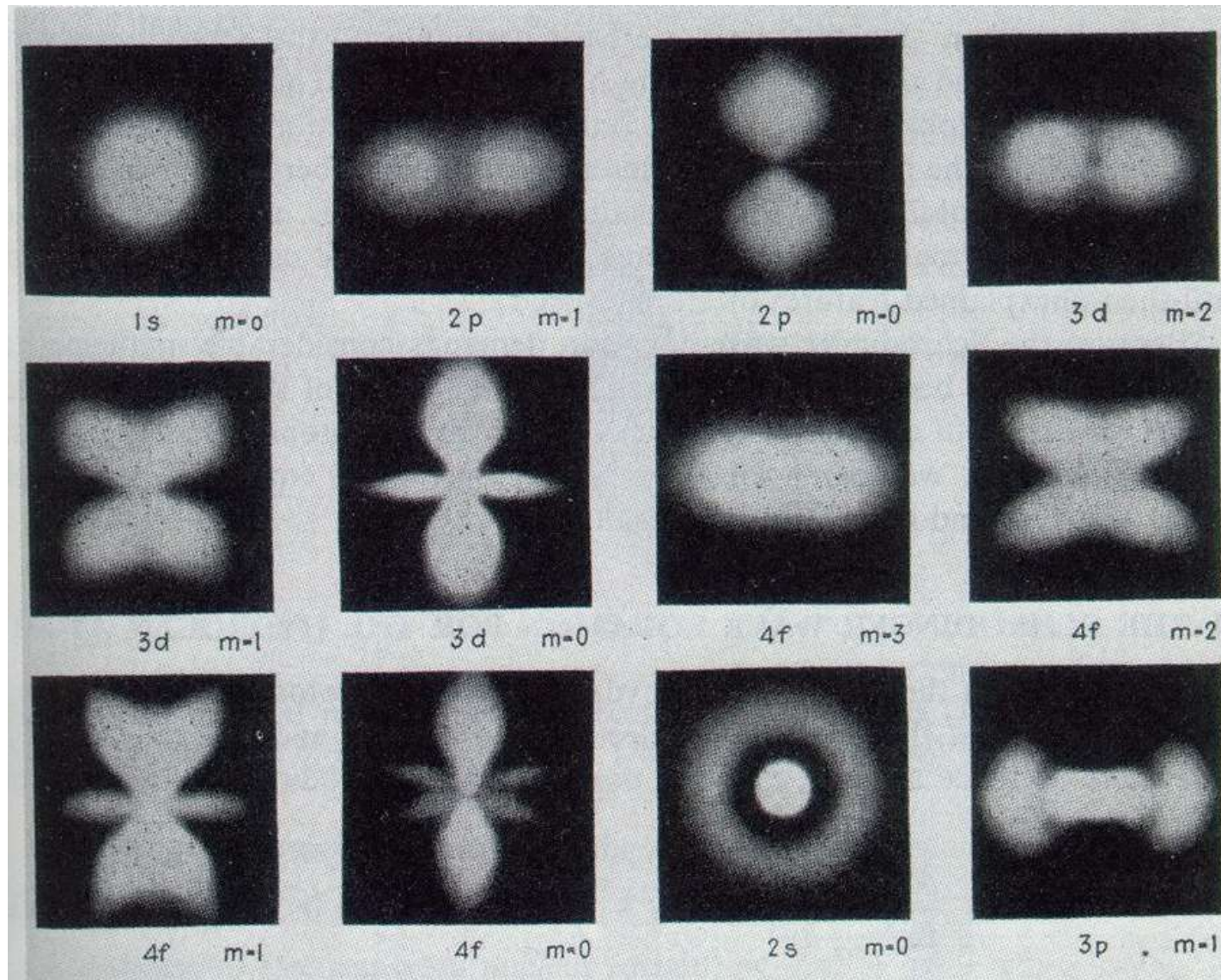
$$\varphi(r, \theta, \phi) = \varphi_{n,l,m_l}(r, \theta, \phi) = R_{n,l}(r) Y_{l,m_l}(\theta, \phi)$$



e Orbital: e Cloud (Simulated)



e Orbital: e Cloud (Experimental)



Quantized Electron Energy

Schrödinger equation $\frac{d^2\varphi(x)}{dx^2} + \frac{2m}{\hbar^2}(E - V)\varphi(x) = 0$

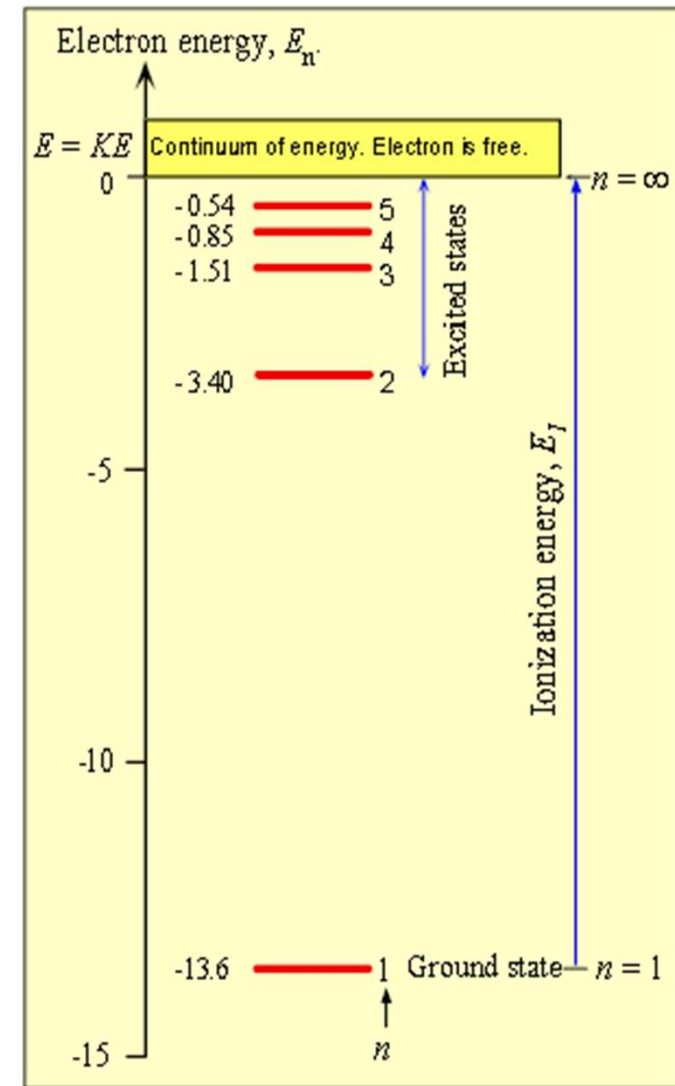
By solving the SWE, we find the E of ‘e’

$$E_n = -\frac{me^4Z^2}{8\varepsilon_0\hbar^2n^2} \quad \text{depends only on “}n\text{”}$$

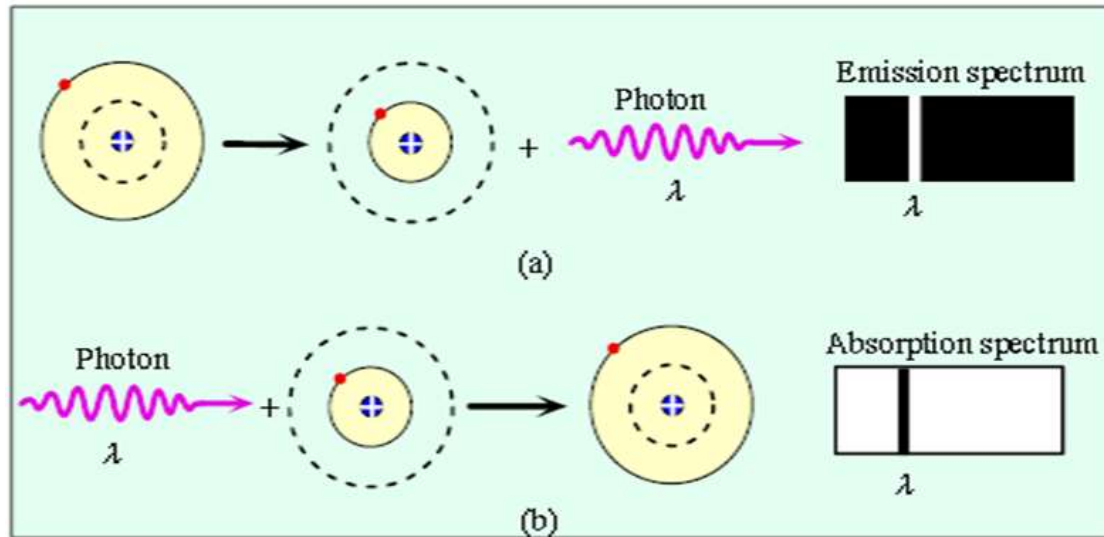
$$E_n = -\frac{Z^2 E_1}{n^2} = -\frac{Z^2 (13.6 \text{ eV})}{n^2}$$

where $E_1 = -\frac{me^4}{8\varepsilon_0\hbar^2} = 2.18 \times 10^{-18} \text{ J} = 13.6 \text{ eV}$

E_1 corresponds to ‘ionization energy’: energy required to remove the electron in the hydrogen atom from the lowest energy level E_1 to infinity.

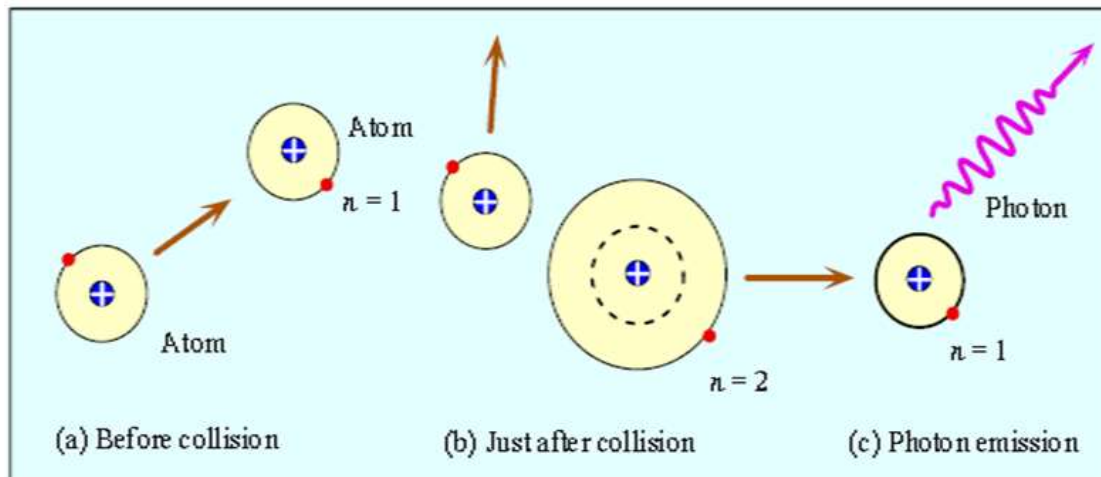


Excited Electron



By Excitation

By virtue of the quantization of E , we see that the emission and absorption of light from excited atoms **can only have certain wavelengths**.



By Atom Collision

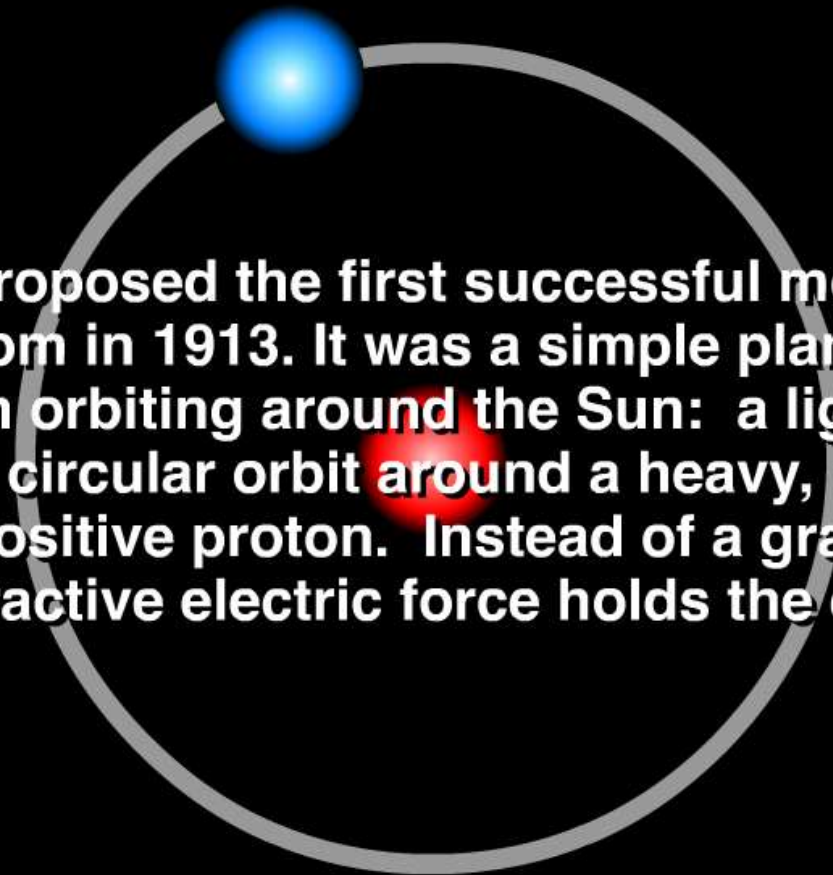
An atom can become excited by a collision with another atom. When it returns to its ground state, it emits a photon.

HeCd/HeNe lasers



Bohr's Atom Model (1913)

The Bohr model of the hydrogen atom

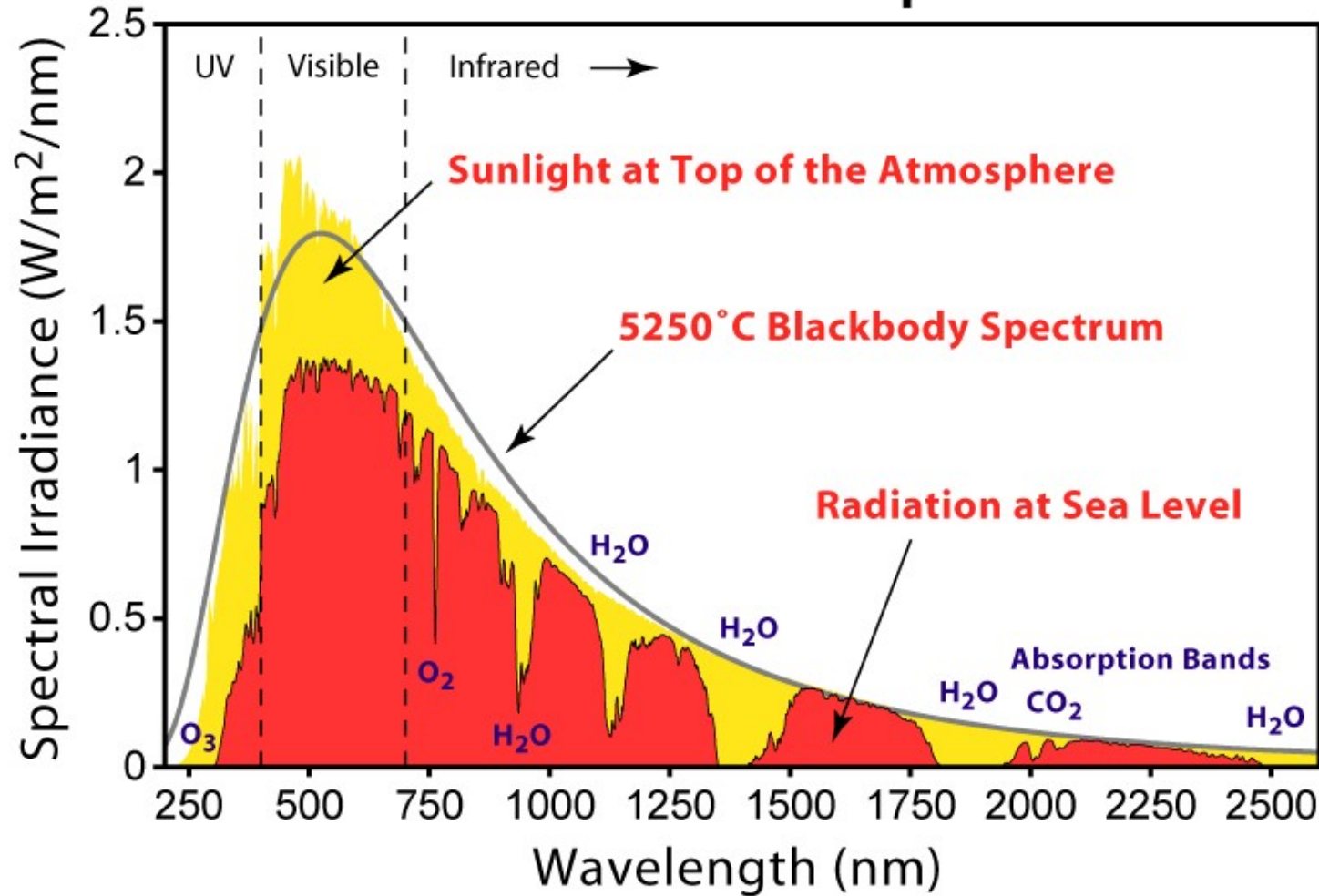


Niels Bohr proposed the first successful model of the hydrogen atom in 1913. It was a simple planetary model like the Earth orbiting around the Sun: a light negative electron in a circular orbit around a heavy, nearly stationary, positive proton. Instead of a gravitational force, an attractive electric force holds the electron in its orbit.

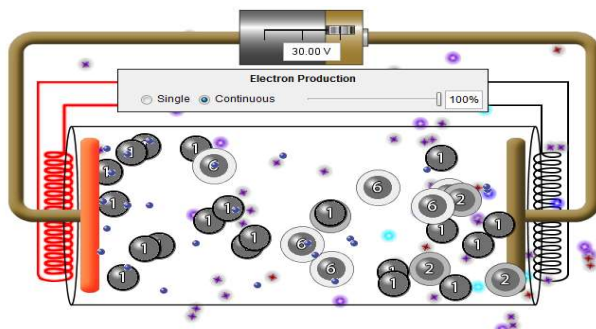
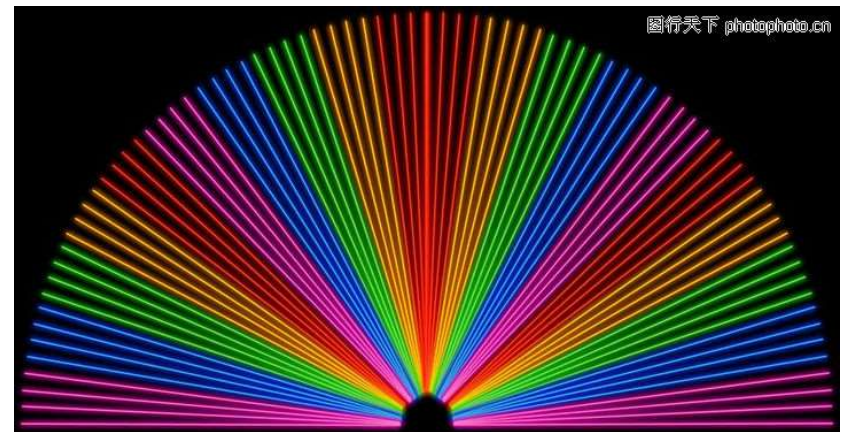


Dark Lines in Spectrum

Solar Radiation Spectrum



Neon Light (Gas Discharging)



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Bohr's Atom Model (1913)



The Nobel Prize in Physics 1922
Niels Bohr

The Nobel Prize in Physics 1922



Niels Henrik David Bohr

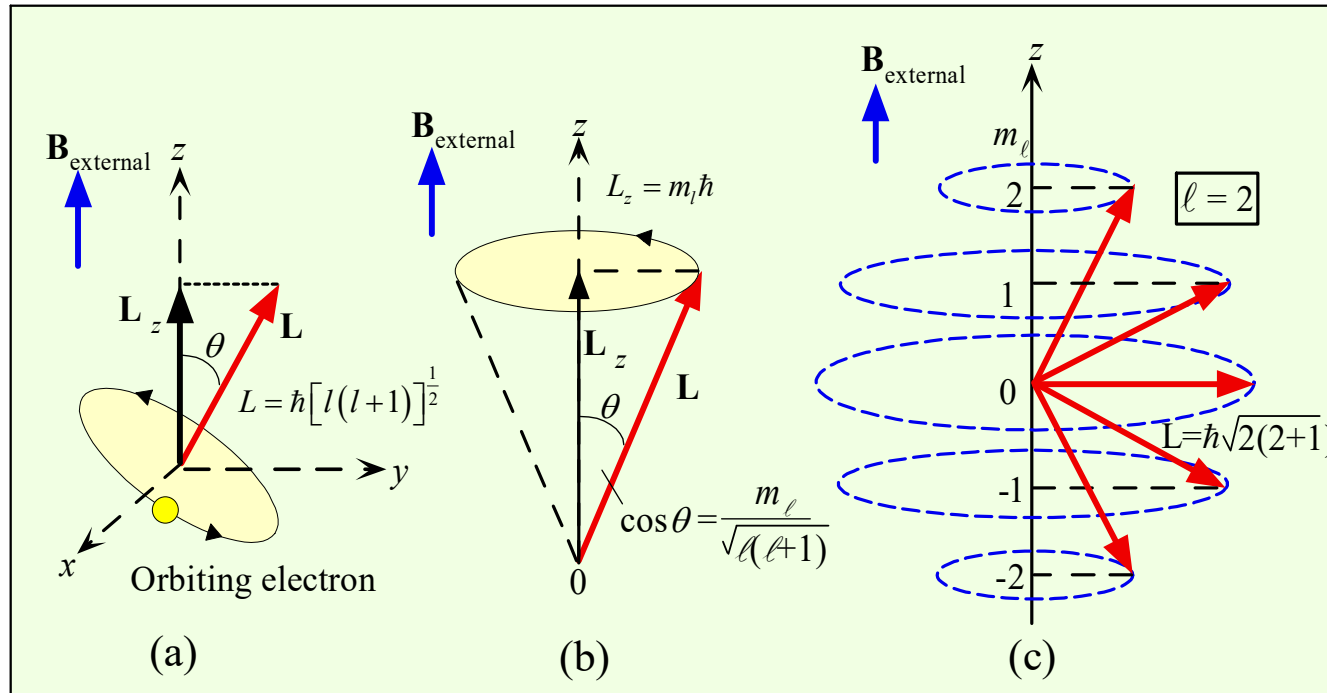


The Nobel Prize in Physics 1922 was awarded to Niels Bohr *"for his services in the investigation of the structure of atoms and of the radiation emanating from them"*.



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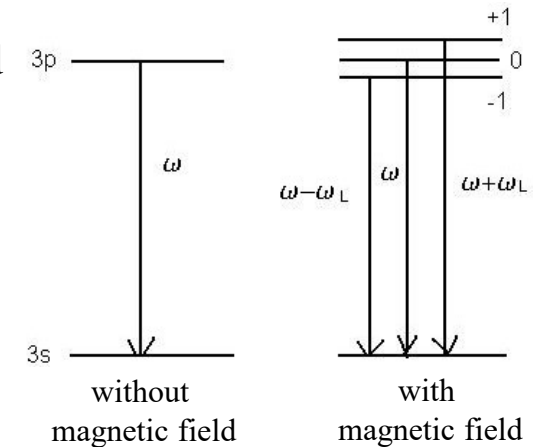
L and m_l



The e in the atom has an orbital angular momentum L since the e is attracted to the nucleus by a central force. In QM, L is quantized by the quantum number l .

(a) The electron has an orbital angular momentum which has a quantized component, L_z , along an external magnetic field, $\mathbf{B}_{\text{external}}$. (b) The orbital angular momentum vector $\mathbf{L}$ rotates about the z -axis. Its component L_z is quantized and therefore the orientation of $\mathbf{L}$, the angle θ , is also quantized. $\mathbf{L}$ traces out a cone. (c) According to quantum mechanics, only certain orientations (θ) for $\mathbf{L}$ are allowed as determined by ℓ and m_ℓ .

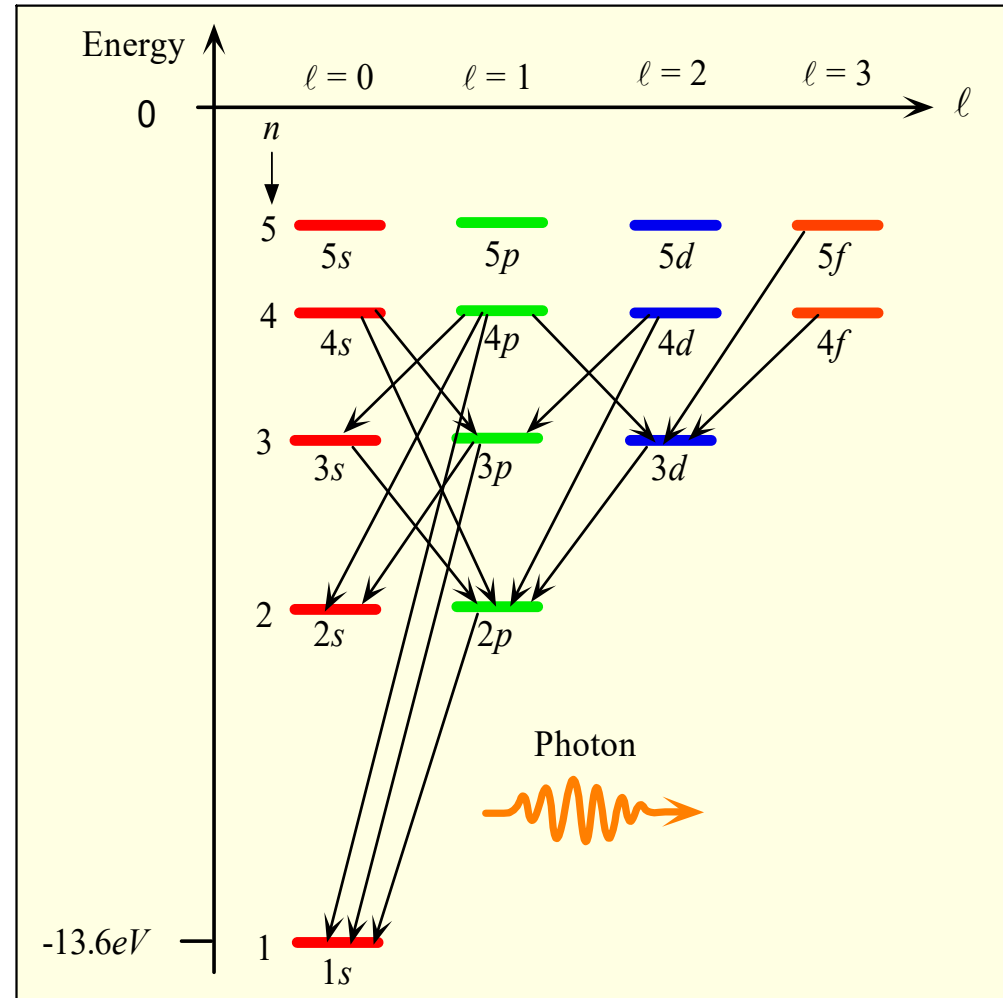
Zeeman Effect



Transition Selection Rules

- An rule that govern which transitions are allowed from one state to another by photon absorption
- Photon has an intrinsic angular momentum with a constant magnitude $\hbar$
- From the law of the **conservation of angular momentum**, photon absorption or emission should obey

$$\Delta l = \pm 1 \quad \text{and} \quad \Delta m_l = 0, \pm 1$$



An illustration of the allowed photon emission processes. Photon emission involves $\Delta \ell = \pm 1$.



Electron Spin

- Electron has a spin angular momentum S due to its rotation about its own axis and S quantized in QM

- In analogy with l and m_l :

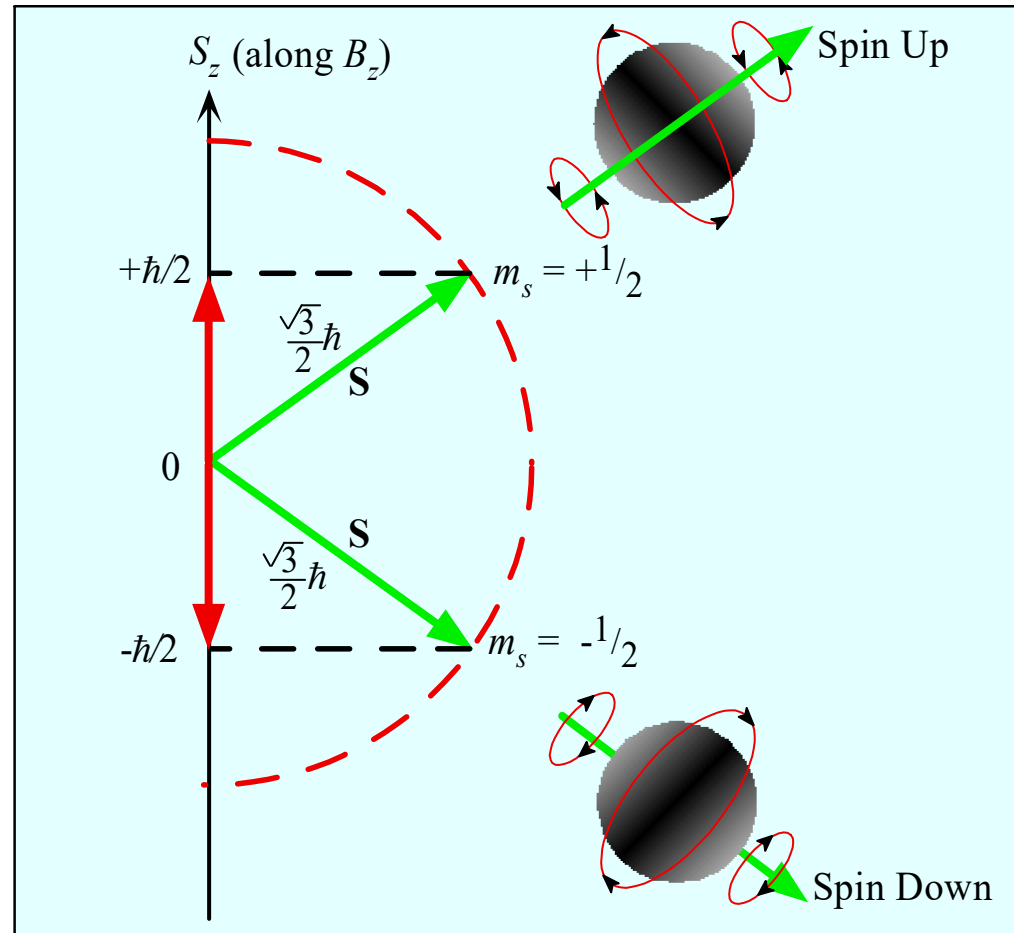
$$L = \hbar [l(l+1)]^{1/2}$$

$$L_z = m_l \hbar$$

intrinsic angular momentum

$$S = \hbar [s(s+1)]^{1/2} = \frac{3}{2}\hbar, \quad S = \frac{1}{2}$$

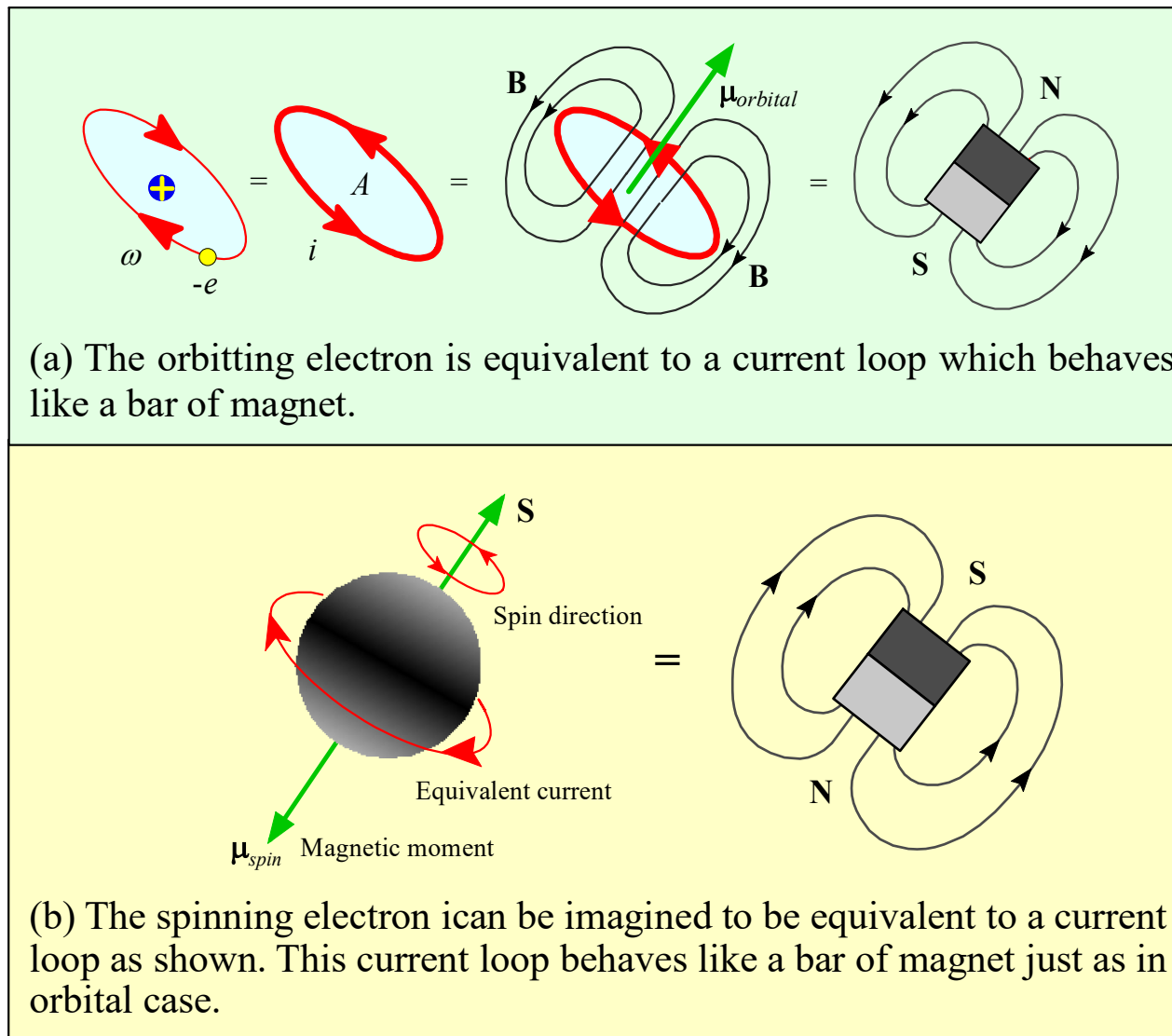
$$S_z = m_s \hbar, \quad m_s = \pm \frac{1}{2}$$



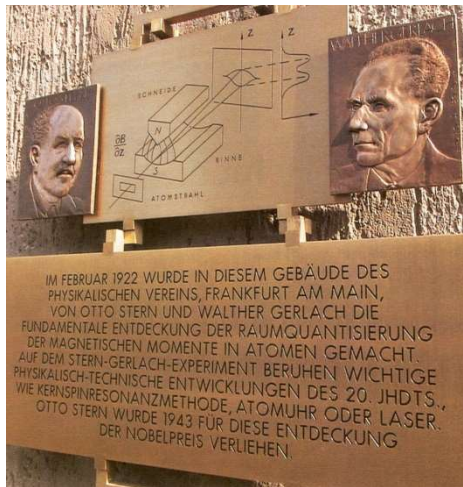
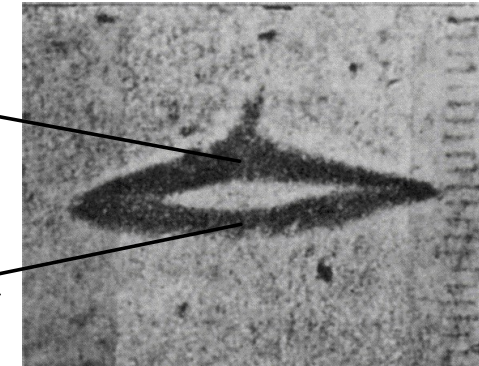
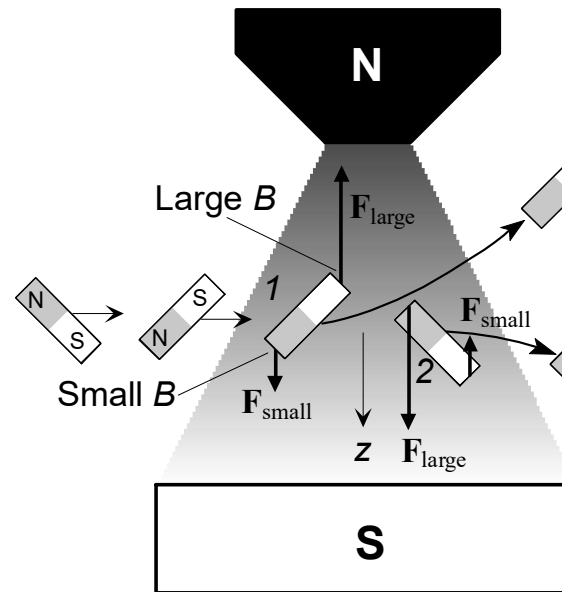
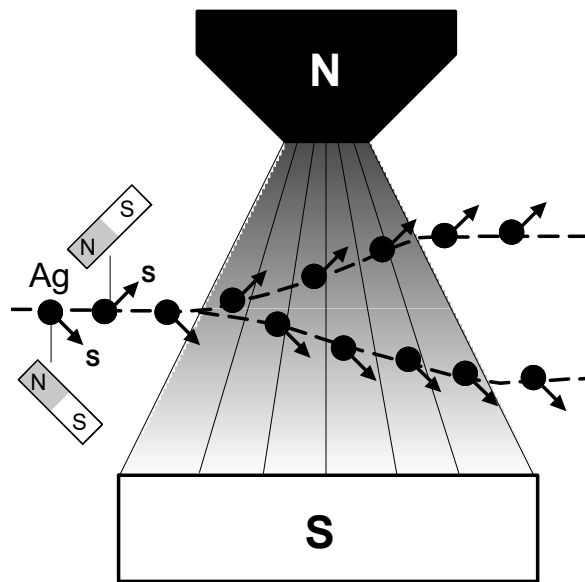
Spin angular momentum exhibits space quantization. Its magnitude along z is quantized so that the angle of $\mathbf{S}$ to the z -axis is also quantized.



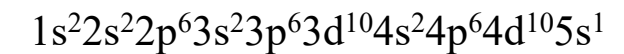
Electron Spin



Stern-Gerlach Experiment



$_{47}\text{Ag}$ 的电子排列方式:



Reading material: Stern and Gerlach: How a bad cigar helped reorient atomic physics



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Stern-Gerlach Experiment



The Nobel Prize in Physics 1943
Otto Stern

The Nobel Prize in Physics 1943



Otto Stern

The Nobel Prize in Physics 1943 was awarded to Otto Stern *"for his contribution to the development of the molecular ray method and his discovery of the magnetic moment of the proton"*.



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